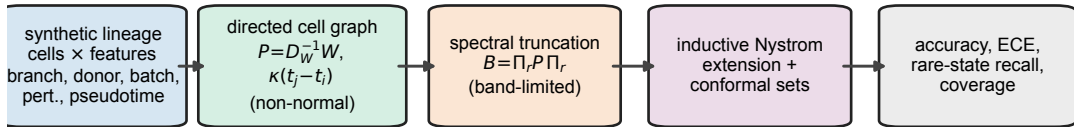


all data synthetic & seeded; every grouping split passes a programmatic no-leakage check



pseudotime-directed (noncommutative) transport × spectral truncation • targets the ~ 4% rare state, no held-out connectivity at inference